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Handling soft constraints with different priorities in explicit model predictive control

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Abstract

In receding horizon model predictive control of discrete-time linear systems it may happen to land on a state which renders the satisfaction of some (soft) constraints impossible. If the controller implements the explicit optimal control law, then the map is not defined at that state and no control action can be applied. This problem can be counteracted relaxing those soft constraints and penalizing the 1-norm of their violation in the objective function. The corresponding solution, as the soft constraint violation is increasingly penalized, converges to a limit piece-wise affine map defined also for infeasible states. The map returns the original optimal control action when the state is feasible, and returns a control action minimizing the soft constraint violation when the state is infeasible. Building on this, in this thesis we show also that is possible to obtain a solution with the same properties but which also takes into account different priority levels among soft constraints. This is possible using different penalty coefficients for constraints with different priority levels, and letting those coefficients go to infinity in a specific order, according to the priority order, to compute the limit map.

Keywords: explicit MPC, soft constraints, constraints prioritization, multi-parametric quadratic programming

Abstract in lingua italiana

Nel controllo predittivo a orizzonte recessivo per sistemi lineari a tempo discreto, può accadere di finire in uno stato che rende impossibile soddisfare alcuni vincoli (soft). Se il controllore implementa la legge di controllo ottimo esplicita, allora la mappa non è definita per quello stato e non può essere applicata alcuna azione di controllo. Questo problema può essere aggirato rilassando tali vincoli soft e penalizzando nella funzione obiettivo la norma 1 della loro violazione. La soluzione corrispondente, all'aumentare della penalizzazione della violazione dei vincoli soft, converge verso una mappa limite affine a tratti, definita anche per stati non ammissibili. La mappa restituisce l'azione di controllo ottima originale quando lo stato è ammissibile e fornisce un'azione di controllo che minimizza la violazione dei vincoli soft quando lo stato non è ammissibile. Basandosi su questa metodologia, in questa tesi mostriamo che è possibile ottenere una soluzione con le stesse proprietà, ma che tiene anche conto di diversi livelli di priorità tra i vincoli soft. Ciò è possibile utilizzando coefficienti di penalità differenti per vincoli con diversi livelli di priorità e facendo tendere questi coefficienti all'infinito in un ordine specifico, secondo l'ordine di priorità, per calcolare la mappa limite.

Parole chiave: controllo predittivo esplicito, vincoli rilassati, prioritizzazione dei vincoli, programmazione quadratica multi-parametrica

Notation

- $\mathbb{R}$: the set of real numbers.
- $\mathbb{R}_+$: the set of non-negative real numbers.
- $M \in \mathbb{R}^{n_r, n_c}$: set of matrices with n_r rows and n_c columns.
- $m^{[i]}$: i th row of matrix M .
- M^I : submatrix containing the rows of M indexed by $I \subseteq \{1, \dots, n_r\}$.
- $M^{[:,1:n]}$: submatrix containing all the rows of M and the first n columns.
- $M^{[1:n,:]}$: submatrix containing all the columns of M and the first n rows.
- $M^{[:,i]}$: i th column of matrix M
- M^T : the transpose of M .
- $M \succ 0$: positive definite matrix.
- $1_n/0_n$: n -dimensional vector containing all ones/zeros (the subscript will be omitted when clear from the context).
- $1_{n,m}/0_{n,m}$: the n -by- m one/zero matrix.
- $\|v\|_1$: the 1-norm of vector $v \in \mathbb{R}^n$.
- $[v]_+$: is equal to $\max\{v, 0_n\}$, where the maximum between two vectors has to be intended as component-wise.

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1 | Introduction

1.1. Motivation

The possibility to describe most of physical systems (but also economic, chemical, etc...) through mathematical models allows one to understand in which way it is possible to affect their behaviour in terms of inputs and outputs. Unfortunately, these mathematical representations cannot take into account all the dynamics involved in real systems, which are highly complex.

The control theory exploits the knowledge of the measure (feedback) of the real system state (output) in order to fill the gap with its mathematical representation. The measure of the output is compared with the expected one and the difference is used to generate the input to control the real system.

When designing a controller the engineer must satisfy some performance requirements asked by the clients. Furthermore, constraints both on the inputs, due to physical limits, and on the states, for keeping desirable functioning conditions, must also be considered. These last considerations motivated the development of a branch of control theory known as optimal control, in which, given the current state, future control inputs are carefully chosen to satisfy all constraints and maximize performance.

Sometimes it could happen that the system reaches a state from which is not possible to leave without violating some constraints. This brakes the optimal control strategy and let the system evolve in open loop. In an industrial process, for example, this could reduce the rate of production significantly, or bring about safety concerns.

The need to solve this issue inspired different approaches with the specific goal of online recovering the system from a constraint violation. However, not all constraints have the same level of criticality or importance, suggesting that specific strategies on the choice of the constraints to violate should be considered.

The work of this thesis takes inspiration from this last observation, with the aim of investigating on how to add some more sophisticated features to the existing controllers in order to recover from infeasibility by relaxing some constraints based on priority levels.

1.2. Optimal Control

Let k be a discrete-time index and let us consider the following discrete-time linear time-invariant (LTI) dynamical system

$$x_{k+1} = Ax_k + Bu_k, \quad (1.1)$$

with $A \in \mathbb{R}^{n,n}$, $B \in \mathbb{R}^{n,m}$, $x_k \in \mathbb{R}^n$ and $u_k \in \mathbb{R}^m$. Let's now suppose we would like to control this system so that, given an initial state x_{k_0} , it converges to the origin as fast as possible while limiting the overall actuation effort. This goal can be achieved by minimizing an *objective function* defined as

$$J_N^*(x_{k_0}, k_0) = \min_{u_{k_0}, u_{k_0+1}, \dots} J_N(x_{k_0}, k_0) = \min_{u_{k_0}, u_{k_0+1}, \dots} \sum_{k=k_0}^{\infty} x_{k+1}^T Q x_{k+1} + u_k^T R u_k, \quad (1.2)$$

where the state and input *penalty weights* are symmetric positive-definite matrices $0 \prec Q^T = Q \in \mathbb{R}^{n,n}$ and $0 \prec R^T = R \in \mathbb{R}^{m,m}$. This control problem is known as the *infinite horizon optimal control problem* and can be solved analytically exploiting the *Bellman's principle of optimality* (Bellman 2010). The resulting optimal control policy is given by $u_k = K^* x_k$ and the control strategy is referred to as *linear quadratic control* (see Anderson and Moore 2007).

1.3. Model Predictive Control

In real world applications, the inputs of the system in (1.1) must satisfy specific technical limits, depending, e.g., on the maximal deliverable power of the actuators. Furthermore, in order to either improve the process performances or avoid critical values for the state of the controlled system, it could be desirable to keep the state of the system within a certain region $X \subseteq \mathbb{R}^n$.

Unfortunately, adding constraints on the inputs or the states in (1.2) prevents us to find an analytical solution and the fact that (1.2) has an infinite number of decision variables $u_{k_0}, u_{k_0+1}, \dots$ prevents us to find even a numerical solution. We thus typically aim for a sub-optimal solution by approximating the infinite horizon cost in (1.2) with a finite horizon one, comprising of the first N terms. This way it is possible to impose linear constraints on both the states and the inputs like $x_k \in X \subseteq \mathbb{R}^n$ and $u_k \in U \subseteq \mathbb{R}^m$ for each k , with X and U typically being sets containing the origin and defined by a finite number of linear inequalities, i.e. $Y = \{y \mid a_j^T y \leq b_j, j = 1, \dots, n_c\}$ with $a_j \in \mathbb{R}^n, a_j \neq 0$

and $y \in \mathbb{R}^n$. The resulting *finite horizon optimal control problem* can be formulated as

$$\begin{aligned}
 J_N^*(x_k, k) = \min_{u_k, \dots, u_{k+N-1}} & \sum_{t=k}^{k+N-1} u_t^T R u_t + x_{t+1}^T Q x_{t+1} \\
 \text{s.t.: } & x_t \in X & t = k+1, \dots, k+N \\
 & u_t \in U & t = k, \dots, k+N-1 \\
 & x_{t+1} = A x_t + B u_t & t = k, \dots, k+N-1,
 \end{aligned} \tag{1.3}$$

which is a *quadratic program* (QP) and can be easily solved.

A good approximation of the infinite horizon optimal controller can be obtained by resorting to the so called *Receding Horizon* (RH) principle: at each time instant k the state x_k is measured, the corresponding QP in (1.3) is solved and the first value of the optimal input sequence $\{u_k^*, \dots, u_{k+N-1}^*\}$ of (1.3) is applied to the system. This control strategy is known as *Model Predictive Controller* (MPC) (cfr. Schwenzer et al. 2021).

1.4. Explicit Model Predictive Control

At the beginning of the 21st century, there persisted an argument (cfr. Bemporad et al. (2000)) that MPC's utility was confined to slow or diminutive systems, due to the computational effort required by MPC online optimization. This inconvenient brought researchers to develop a new strategy in the use of MPC. In particular they noticed that treating the current state x_k as a parameter, problem (1.3) can be regarded as a multi-parametric quadratic problem (mp-QP), whose solution is a piece-wise affine map between the states and the inputs. This map can be computed offline and then used online to control the system in a receding horizon fashion with no need to solve a QP at each new time step k . The resulting control strategy is known as *Explicit* MPC and its great advantage is the possibility to be ran at faster rates in order to deal with real-time constraints.

1.5. Feasibility in MPC

Contrary to the (theoretical) solution to the infinite horizon constrained control problem, with the standard MPC-RH scheme (either the implicit or the explicit version) we cannot guarantee that the system will satisfy actuation and state constraints indefinitely. Consequently it may happen that the system lands onto a state for which it is not possible to find a solution to (1.3). When this happens the problem is deemed infeasible and no

control action can be actuated, effectively leaving the system uncontrolled.

Different solutions have been proposed in the literature to recover from infeasibility. For example, Falsone et al. 2023 relaxes the states constraints and penalizes their violation in the objective function of the mp-QP.

1.6. Prioritized Constraints

While relaxing state constraints seems an intuitive solution to counter infeasibility, not all state constraints have the same level of importance. For example, during an industrial process one could consider more important to respect some safety limits on variables like the temperature, rather than performance constraints as the energy consumption. One solution could be giving these constraints different priorities: if the system is forced to violate one constraint, then we would like the solver to return the solution which violates the constraints with lower priority.

In the literature there are different approaches to constraints prioritization. In Vada et al. (1999), constraints are divided into hard and soft (relaxed) and then grouped in a hierarchical number of sets ordered according to their priority. When the standard optimization problem of MPC is not feasible, the algorithm starts solving a sequence of QP problems. The first iteration of the algorithm tries to find a solution which must satisfy only the set of constraints with the highest priority. At each new iteration, a new set of constraints according to its priority is added to the QP, until which a solution cannot be found. At this point the last set of constraints added is relaxed with a slack variable computed using the *minimal time approach* (see Rawlings and Muske (1993)). Doing so, the current QP turns out to be feasible with an optimal minimal violation of the last constraints added. The same procedure is then applied iteratively for each remaining soft constraint, so that the violations are minimized sequentially, ensuring therefore that the priority of the constraints is respected. However, this procedure suffers from high computational demands because the number of QP problems to solve increases as the number of constraints grows.

An alternative framework, more general and flexible, is presented in Kerrigan and Maciejowski (2002). The main idea of this work is to deal with objectives with different priorities formulating a *multi-objective problem*. The solution is obtained solving a hierarchical list of single-objective optimization problems: the one with the highest priority is solved first and its solution adds additional constraints to the next one, and so on. The authors show also how this framework can be implemented to MPC. The main advantage of this approach, or the one in Vada et al. 1999, is the possibility to change online the formulation of the problem. However, this comes at the price of an high online

computational burden.

The idea behind the approach in Falsone et al. 2023 is appealing because it deals with the parametric solution (which has a low computational effort online), but all constraints are penalized in the same way and no prioritization is considered.

1.7. Goal

The goal of this thesis is therefore to extend the constraint relaxation paradigm proposed in Falsone et al. 2023 in order to take into account also the priority among the constraints to be relaxed.

2 | Explicit MPC with Soft Constraints

We dedicate this Chapter to illustrate the main results achieved in Falsone et al. (2023), showing that, in the context of constrained optimal control problems solved via explicit model predictive control, it is possible to guarantee applicability of the control action exploiting a specific constraint relaxation strategy.

In the following we make a distinction between:

- *hard constraints*: these constraints cannot be violated. They are usually either the technical limits of the inputs variables, or some critical state variables boundaries to not be crossed;
- *soft constraints*: these can be violated and they are usually related to system performances or desired operating conditions.

2.1. Formulation of the Relaxed Problem

Let's consider problem formulation (1.3) and substitute the Lagrange formula of system (1.1) for $x_{k+i} = A^i x_k + \sum_{j=0}^{i-1} A^j B u_{k+i-1-j}$. We also indicate with $z = [u_k^T \cdots u_{k+N-1}^T]^T \in \mathbb{R}^{n_z}$, $n_z = Nm$, the vector of decision variables and with $\vartheta = x_k \in \mathbb{R}^{n_\vartheta}$, where $n_\vartheta = n$, the parameter vector. If X and U are polytopic sets, then performing some algebraic computations, it is possible to reformulate problem (1.3) as the following mp-QP

$$\begin{aligned}
 \min_z \quad & \frac{1}{2} z^T H z + (F \vartheta + c)^T z \\
 \text{s.t:} \quad & G_h z \leq b_h + S_h \vartheta \\
 & G_s z \leq b_s + S_s \vartheta
 \end{aligned} \tag{2.1}$$

where $0 \prec H = H^T \in \mathbb{R}^{n_z, n_z}$, $F \in \mathbb{R}^{n_z, n_\vartheta}$ and $c \in \mathbb{R}^{n_z}$ define the cost function, and $G_h \in \mathbb{R}^{n_h, n_z}$, $G_s \in \mathbb{R}^{n_s, n_z}$, $b_h \in \mathbb{R}^{n_h}$, $b_s \in \mathbb{R}^{n_s}$, $S_h \in \mathbb{R}^{n_h, n_\vartheta}$, and $S_s \in \mathbb{R}^{n_s, n_\vartheta}$ define the $n_c = n_h + n_s$ constraints, which are divided into in hard constraints (subscript h) and soft constraints (subscript s).

Consider now the following assumption:

Assumption 2.1 (Falsone et al. 2023). The set $\Theta_f \subseteq \mathbb{R}^{n_\vartheta}$ of ϑ for which (2.1) is feasible is full dimensional. Moreover, the constraint set $\mathcal{Z}_\vartheta = \{z \in \mathbb{R}^{n_z} : G_h z \leq b_h + S_h \vartheta\}$ is bounded for some parameter value $\vartheta \in \mathbb{R}^{n_\vartheta}$.

Note that this assumption is naturally satisfied in standard MPC problems, where the evolution of the state ϑ is full dimensional and the actuation constraints, which are independent of ϑ , forces the input $z \in \mathcal{Z}_\vartheta$ to be bounded.

Under Assumption 2.1, problem (2.1) admits an optimal solution given by a Piece-Wise Affine (PWA) function $z^* : \Theta_f \rightarrow \mathcal{Z}$, $\mathcal{Z} = \bigcup_{\vartheta \in \mathbb{R}^{n_\vartheta}} \mathcal{Z}_\vartheta$, like

$$z^*(\vartheta) = K_z^i \vartheta + k_z^i, \quad \vartheta \in \Theta_i, \quad i = 1, \dots, r, \quad (2.2)$$

where $\{\Theta_i\}_{i=1}^r$ is a polyhedral partition of Θ_f , which is also polyhedral (cfr. Bemporad et al. (2000: Section 3)).

Following Falsone et al. 2023 the relaxed version of (2.1) is given by

$$\begin{aligned} \min_z \quad & \frac{1}{2} z^T H z + (F \vartheta + c)^T z + \mu \| [G_s z - b_s - S_s \vartheta]_+ \|_1 \\ \text{s.t:} \quad & G_h z \leq b_h + S_h \vartheta. \end{aligned} \quad (2.3)$$

Notice that, in this relaxed version, all soft constraints disappeared, while their violation is taken into account and penalized as 1-norm in the cost function with a scalar penalty coefficient $\mu > 0$. For a fixed μ the solution of this new mp-QP would be the one trading between the minimization of the 1-norm violation of the soft constraints and the minimization of the original objective function. Under Assumption 2.1, problem (2.3) is feasible for all values of $\vartheta \in \Theta$, where $\Theta = \{\vartheta \in \mathbb{R}^{n_\vartheta} : \mathcal{Z}_\vartheta \neq \emptyset\}$ is such that $\Theta_f \subseteq \Theta \subseteq \mathbb{R}^{n_\vartheta}$.

While it seems that the presence of the operator $[\cdot]_+$ makes the optimization problem non-linear and non-quadratic (thus making the parametric problem more difficult to solve), Falsone et al. (2023: Lemma 1) states that using a vector of auxiliary variables $h \in \mathbb{R}^{n_s}$ problem (2.3) can be reformulated into a problem whose solution (z, h) admits the same optimal z values of the original one (2.3).

The reformulated problem reads as

$$\begin{aligned}
\min_{z,h} \quad & \frac{1}{2} z^T H z + (F_0 p + c)^T z + p^T I_0 h \\
\text{s.t:} \quad & G_h z \leq b_h + S_{h0} p \\
& G_s z \leq b_s + S_{s0} p + h \\
& h \geq 0,
\end{aligned} \tag{2.4}$$

where $F_0 = [F \ 0_{n_z}]$, $I_0 = [0_{n_s, n_\vartheta} \ 1_{n_s}]^T$, $S_{h0} = [S_h \ 0_{n_h}]$, $S_{s0} = [S_s \ 0_{n_s}]$ and $p = [\vartheta^T \ \mu]^T$ being the new parameter vector. Problem (2.4) is again a multi-parametric quadratic problem and hence by Falsone et al. (2023: Corollary 1) problem (2.4), and thus problem (2.3) for Falsone et al. (2023: Lemma 1), admits an optimal solution $z : \Theta \times \mathbb{R}_+ \rightarrow \mathcal{Z}$ which is a PWA map defined over a finite polyhedral partition of $\Theta \times \mathbb{R}_+$.

2.2. Analysis of the Relaxed Problem

2.2.1. Preliminary Concepts

Let's first recall for the sake of clarity some notions from Rockafellar 1970 in the field of convex analysis that are used in the sequel.

Definition 2.1 (Rockafellar 1970: page 61). Let C be a non-empty convex set in $\mathbb{R}^n$. We say that C recedes in the direction d if C includes all the half-lines along direction d which start at points in C . Formally, C recedes along direction y , where $y \neq 0$, if and only if $(x + \lambda y) \in C$ for every $\lambda \in \mathbb{R}_+$ and $x \in C$. The set of all vectors $y \in \mathbb{R}^n$ satisfying the former condition, including $y = 0$, is called the *recession cone* of C . A direction in which C recedes will be referred to as *direction of recession* of C .

2.2.2. A Limit Map

Given that μ trades between the cost and the constraint violation, it is natural to ask how the solution of problem (2.3) behaves as μ grows. According to Falsone et al. (2023: Theorem 1), the optimal map $z : \Theta \times \mathbb{R}_+ \rightarrow \mathcal{Z}$ of (2.3) satisfies

$$\lim_{\mu \rightarrow \infty} z(\vartheta, \mu) = \bar{z}(\vartheta), \tag{2.5}$$

where $\bar{z} : \Theta \rightarrow \mathcal{Z}$ is a PWA function defined over a polyhedral partition $\{\Theta_i\}_{i=1}^q$ of Θ , which is continuous whenever $z(\vartheta, \mu)$ is continuous. Falsone et al. 2023 also precisely

characterize $\bar{z}$. Each region $\Theta_i = \{\vartheta \in \Theta : F_i^0 \vartheta \leq g_i^0\}$ of map $\bar{z}$ can be obtained from a region $\mathcal{R}_i = \{(\vartheta, \mu) \in \Theta \times \mathbb{R}_+ : F_i \vartheta + f_i \mu \leq g_i\}$ of map z receding along the μ -axis (or, equivalently, such that $f_i \leq 0$), setting F_i^0 and g_i^0 equal to the rows of F_i and g_i corresponding to zero elements of f_i .

Moreover, for each polyhedral region Θ_i , with $i = 1, \dots, q$, the affine map

$$\bar{z}(\vartheta) = K_z^i \vartheta + k_z^i, \quad \vartheta \in \Theta_i, \quad (2.6)$$

is derived from the affine map $z(\vartheta, \mu) = K_z^i \vartheta + M_z^i \mu + k_z^i$ defined over the corresponding region $\mathcal{R}_i$.

2.2.3. Relation with the Original Non-Relaxed Problem

According to Falsone et al. (2023: Theorem 2), under [Assumption 2.1](#), the limit map $\bar{z}$ in (2.5) satisfies

$$\begin{aligned} \bar{z}(\vartheta) \in \arg \min_{z, h} \quad & \frac{1}{2} z^T H z + (F \vartheta + c)^T z \\ \text{s.t.:} \quad & G_h z \leq b_h + S_h \vartheta \\ & \|[G_s z \leq b_s + S_s \vartheta]_+\|_1 \leq \bar{v}(\vartheta), \end{aligned} \quad (2.7)$$

where the soft constraints are relaxed up to a 1-norm violation of

$$\begin{aligned} \bar{v}(\vartheta) = \min_{z, h} \quad & \|[G_s z \leq b_s + S_s \vartheta]_+\|_1 \\ \text{s.t.:} \quad & G_h z \leq b_h + S_h \vartheta, \end{aligned} \quad (2.8)$$

which is the minimum 1-norm violation of the soft constraints that can be obtained enforcing the hard ones. Accordingly, solving (2.3) as $\mu \rightarrow \infty$ is equivalent to find the best map $\bar{z}(\vartheta)$ such that:

- it minimizes the 1-norm of the soft constraints violation, as a primary (higher priority) objective, and, then,
- among those solutions achieving the minimum 1-norm violation, it selects the one that optimize the original objective function.

This last considerations lead to another important result. Since $\vartheta \in \Theta_f$ implies that both soft and hard constraints can be satisfied, then it also implies $\bar{v}(\vartheta) = 0$ in (2.7), meaning that problem (2.7) reduces to the original mp-QP in (2.1). This further implies

$$\bar{z}(\vartheta) = z^*(\vartheta), \quad \vartheta \in \Theta_f, \quad (2.9)$$

as stated in Falsone et al. (2023: Corollary 2).

This is an important result because it proves that if no violation is required the solution of the relaxed problem is the same of the original one.

2.3. Solution of the Relaxed Problem

Falsone et al. (2023: Algorithm 1) also provides a method to actually find the optimal limit map, which will, by construction satisfy (2.7) and, in particular, (2.9).

Algorithm 2.1: Procedure to compute $\bar{z}(\vartheta)$

- 1: $\mathcal{P}_r \leftarrow$ optimal map $\bar{z}(\vartheta, \mu)$ of (2.3)
 - 2: $\mathcal{P} \leftarrow \emptyset$
 - 3: **for all** $(F_i, f_i, g_i, K_z^i, M_z^i, k_z^i) \in \mathcal{P}_r$ **do**
 - 4: **if** $f_i \leq 0$ **then**
 - 5: $\mathcal{O} \leftarrow \{j : f_i^{[j]} = 0\}$
 - 6: $F_i^0 = F_i^{\mathcal{O}}$ and $g_i^0 = g_i^{\mathcal{O}}$
 - 7: $\mathcal{P} \leftarrow \mathcal{P} \cup (F_i^0, g_i^0, K_z^i, k_z^i)$
 - 8: **end if**
 - 9: **end for**
-

First, in Step 1 the optimal map $z(\vartheta, \mu)$ of (2.3) (or equivalently (2.4)) is computed and stored as a set $\mathcal{P}_r$ of 6-tuples, where each i th element $(F_i, f_i, g_i, K_z^i, M_z^i, k_z^i) \in \mathcal{P}_r$ encodes the fact that $z(\vartheta, \mu) = K_z^i \vartheta + M_z^i \mu + k_z^i$ for all (ϑ, μ) such that $F_i \vartheta + f_i \mu \leq g_i$. In Step 2 a new empty set $\mathcal{P}$ is initialized for encoding the optimal map $\bar{z}(\vartheta)$. Then, for each element in $\mathcal{P}_r$ (cfr. Step 3) it is checked if the positive μ -axis is a direction of recession for the corresponding region $\{(\vartheta, \mu) : F_i \vartheta + f_i \mu \leq g_i\}$ (cfr. Step 4). If the condition is satisfied, the constraints that do not depend on μ defining the current region are identified (cfr. Step 5), are stored in g_i^0 and F_i^0 (cfr. Step 6) and, then, the 4-tuple $(F_i^0, g_i^0, K_z^i, k_z^i)$ is added to $\mathcal{P}$ (cfr. Step 7), encoding the fact that $\bar{z}(\vartheta) = K_z^i \vartheta + k_z^i$ for all $\vartheta \in \Theta_i = \{\vartheta : F_i^0 \vartheta \leq g_i^0\}$.

2.4. Summary

In this chapter we have discussed how infeasibility in mp-QP problems can be handled resorting to the constraint relaxation technique described in Falsone et al. 2023. In order to do so, we have formulated the relaxed version of the original hard constrained problem where the 1-norm violation of the soft constraints is penalized through a coefficient μ in the objective function, and illustrated the existence of a limit map $\bar{z}(\vartheta)$ which coincides as

$\mu \rightarrow \infty$ with the optimal map $z^*(\vartheta)$ of the original non-relaxed problem, when the current state makes the original problem feasible, otherwise it minimizes the soft constraints violation first and, then, the original objective function.

The next goal will be to find a new formulation of the problem which, besides relaxing constraints, will try to take into account also their priority level, trying not to violate those constraints which are considered more critical than others.

3 | Explicit MPC with prioritised constraints

This chapter describes how different priorities among soft constraints can be accounted for in explicit MPC. We will extend the approach revised in the previous chapter, to account for different level of priorities and we will conjecture how the properties of the relaxed problem extend to this setup.

3.1. Formulation of the Relaxed Problem with Prioritized Constraints

Let us consider the following mp-QP

$$\begin{aligned}
 \min_{z,h} \quad & \frac{1}{2}z^T H z + (F\vartheta + c)^T z \\
 \text{s.t:} \quad & G_h z \leq b_h + S_h \vartheta \\
 & G_s^1 z \leq b_s^1 + S_s^1 \vartheta \\
 & \vdots \\
 & G_s^{n_p} z \leq b_s^{n_p} + S_s^{n_p} \vartheta,
 \end{aligned} \tag{3.1}$$

where the soft constraints with priority level i are denoted as

$$G_s^i z \leq b_s^i + S_s^i \vartheta,$$

such that, if n_s^i is the number of constraints inside the i th set, then $G_s^i \in \mathbb{R}^{n_s^i, n_z}$, $b_s^i \in \mathbb{R}^{n_s^i}$ and $S_s^i \in \mathbb{R}^{n_s^i, n_\vartheta}$. The constraints labeled with a higher i are the ones with higher priority: $i = n_p$ indicates the highest priority, $i = n_p - 1$ the second priority and so on. The idea is to relax the constraints as done in the previous chapter but respecting the order of priorities.

Similarly to the previous chapter we make the following assumption

Assumption 3.1. The set $\Theta_f \subseteq \mathbb{R}^{n_\vartheta}$ of ϑ for which (3.1) is feasible is full dimensional. Moreover, the constraint set $\mathcal{Z}_\vartheta = \{z \in \mathbb{R}^{n_z} : G_h z \leq b_h + S_h \vartheta\}$ is bounded for some parameter value $\vartheta \in \mathbb{R}^{n_\vartheta}$.

Inspired by Falsone et al. 2023 the relaxed version of (3.1) is given by

$$\begin{aligned} \min_z \quad & \frac{1}{2} z^T H z + (F\vartheta + c)^T z + \sum_{i=1}^{n_p} \mu_i \| [G_s^i z - b_s^i - S_s^i \vartheta]_+ \|_1 \\ \text{s.t:} \quad & G_h z \leq b_h + S_h \vartheta \end{aligned} \quad (3.2)$$

where $\mu_i \in \mathbb{R}_+$ is the penalty coefficient related to constraints with priority level $i = 1, \dots, n_p$, being n_p the number of priority levels considered. Under Assumption 2.1, problem (3.2) is feasible for all values of $\vartheta \in \Theta$, where $\Theta = \{\vartheta \in \mathbb{R}^{n_\vartheta} : \mathcal{Z}_\vartheta \neq \emptyset\}$ is such that $\Theta_f \subseteq \Theta \subseteq \mathbb{R}^{n_\vartheta}$.

It can be noticed that this formulation is very similar to problem (2.3), the main difference being the presence of a different penalty for each set of constraints with a different priority level.

In order to get rid of the presence of the non-linear operator $[\cdot]_+$, we write the following new problem

$$\begin{aligned} \min_{z, h} \quad & \frac{1}{2} z^T H z + (F\vartheta + c)^T z + \sum_{i=1}^{n_p} \mu_i 1^T h_i \\ \text{s.t:} \quad & G_h z \leq b_h + S_h \vartheta \\ & G_s^1 z \leq b_s^1 + S_s^1 \vartheta + h_1 \\ & \vdots \\ & G_s^{n_p} z \leq b_s^{n_p} + S_s^{n_p} \vartheta + h_{n_p} \\ & h_1, \dots, h_{n_p} \geq 0, \end{aligned} \quad (3.3)$$

where each $h_i \in \mathbb{R}^{n_s^i}$ is the vector of auxiliary variables whose dimension is equal to the number of n_s^i constraints with priority i . Conjecturing an extension of Falsone et al. (2023: Lemma 1) we guess that problem (3.3) admits as solution a pair $(z(\vartheta, \boldsymbol{\mu}), h(\vartheta, \boldsymbol{\mu}))$ such that $z(\vartheta, \boldsymbol{\mu})$ is also optimal for (3.2), where $\boldsymbol{\mu} = [\mu_1 \cdots \mu_{n_p}]^T$.

Now using as parameter vector $p = [\vartheta^T \quad \boldsymbol{\mu}^T]^T$, we can rewrite problem (3.3) in the following equivalent form

$$\begin{aligned}
\min_{z,h} \quad & \frac{1}{2} z^T H z + (F_0 p + c)^T z + \sum_{i=1}^{n_p} p^T I_0^i h_i \\
\text{s.t:} \quad & G_h z \leq b_h + S_{h0} p \\
& G_s^1 z \leq b_s^1 + S_{s0}^1 p + h_1 \\
& \vdots \\
& G_s^{n_p} z \leq b_s^{n_p} + S_{s0}^{n_p} p + h_{n_p} \\
& h_1, \dots, h_{n_p} \geq 0,
\end{aligned} \tag{3.4}$$

with $F_0 = [F \ 0_{n_z, n_p}]$, $I_0^i = [0_{n_s^i, n_\vartheta} \ 0_{n_s^i, i-1} \ 1_{n_s^i} \ 0_{n_s^i, n_p-i}]^T$, $S_{h0} = [S_h \ 0_{n_h, n_p}]$, $S_{s0}^i = [S_s^i \ 0_{n_s^i, n_p}]$, and $p \in \mathbb{R}^{n_\vartheta + n_p}$, which is again a multi-parametric quadratic problem and consequently we conjecture that Falsone et al. (2023: Corollary 1) can be extended to show that (3.4), and thus (3.3) and (3.2), admit an optimal solution $z : \Theta \times \mathbb{R}_+^{n_p} \rightarrow \mathcal{Z}$ which is a PWA map defined over a finite polyhedral partition of $\Theta \times \mathbb{R}_+^{n_p}$.

3.2. Analysis of the Relaxed Problem

3.2.1. A New Limit Map

In the previous chapter we have seen how according to Falsone et al. (2023: Theorem 1) it is possible to obtain a limit map $\bar{z} : \Theta \rightarrow \mathcal{Z}$ by letting μ in $z(\vartheta, \mu)$ go to infinity. We conjecture that we can do so also in our case, where we have a vector of weights $\boldsymbol{\mu}$ in the optimal map $z(\vartheta, \boldsymbol{\mu})$ of (3.2).

In order to obtain a limit map that reflects the different priority levels we guess that we need to perform a sequence of limits along each component of $\boldsymbol{\mu}$ approaching infinity. In particular, the order in which this sequence of limits is performed will determine the order in which the constraints are relaxed whenever it is not possible to find a feasible solution for the original problem (3.1).

Let's use $\boldsymbol{\mu}_j$ to denote the vector $[\mu_1 \ \dots \ \mu_{n_p-j}]^T \in \mathbb{R}^{n_p-j}$, for $j = 1, \dots, n_p - 1$. We perform the first limit along μ_{n_p} , which is associated with the highest priority constraints

$$\lim_{\mu_{n_p} \rightarrow \infty} z(\vartheta, \boldsymbol{\mu}) = \bar{z}(\vartheta, \boldsymbol{\mu}_1). \tag{3.5}$$

We expect to obtain, according to Falsone et al. (2023: Theorem 1), a first limit map $\bar{z}(\vartheta, \boldsymbol{\mu}_1)$ defined over a polyhedral partition $\{\mathcal{P}_i\}_{i=1}^{q_1}$ of $\Theta \times \mathbb{R}_+^{n_p-1}$ such that for each

$i = 1 \dots q_1$ we have

$$\bar{z}(\vartheta, \boldsymbol{\mu}_1) = K_i \vartheta + L_i \boldsymbol{\mu}_1 + k_i, \quad (\vartheta, \boldsymbol{\mu}_1) \in \mathcal{P}_i,$$

independent from μ_{n_p} . Performing all the remaining limits in sequence according to the order of priorities we expect to obtain

$$\lim_{\mu_1 \rightarrow \infty} \cdots \lim_{\mu_j \rightarrow \infty} \cdots \lim_{\mu_{n_p} \rightarrow \infty} z(\vartheta, \boldsymbol{\mu}) = \bar{z}(\vartheta), \quad (3.6)$$

where the map $\bar{z} : \Theta \rightarrow \mathcal{Z}$ is a PWA function defined over a polyhedral partition $\{\Theta_i\}_{i=1}^q$ of Θ , such that for each $i = 1, \dots, q$

$$\bar{z}(\vartheta) = K_i \vartheta + k_i, \quad \vartheta \in \Theta_i,$$

is independent from $\boldsymbol{\mu}$, each Θ_i being related to the i th region $\mathcal{R}_i = \{(\vartheta, \mu_1) \in \Theta \times \mathbb{R}_+ : F_i \vartheta + f_i \mu_1 \leq g_i\}$ of $\bar{z}(\vartheta, \mu_1)$ receding along the μ_1 -axis, and is such that $\Theta_i = \{\vartheta \in \Theta : F_i^0 \vartheta \leq g_i^0\}$, with F_i^0 and g_i^0 containing the rows of F_i and g_i corresponding to zero elements of f_i and $\mathcal{R}_i$ being obtained recursively from a region of $z(\vartheta, \boldsymbol{\mu})$.

3.2.2. Relation with the Original Non-Relaxed Problem

We conjecture an extension of the results obtained in Falsone et al. (2023: Theorem 2) saying that, under [Assumption 3.1](#), for all $\vartheta \in \Theta$, any limit map $\bar{z}$ obtained via (3.6) satisfies

$$\begin{aligned} \bar{z}(\vartheta) \in \arg \min_{z, h} \quad & \frac{1}{2} z^T H z + (F \vartheta + c)^T z \\ \text{s.t.:} \quad & G_h z \leq b_h + S_h \vartheta \\ & \|[G_s^{n_p} z \leq b_s^{n_p} + S_s^{n_p} \vartheta]_+\|_1 \leq \bar{v}_{n_p}(\vartheta) \\ & \vdots \\ & \|[G_s^1 z \leq b_s^1 + S_s^1 \vartheta]_+\|_1 \leq \bar{v}_1(\vartheta) \end{aligned} \quad (3.7)$$

with for $i = 1, \dots, n_p$,

$$\begin{aligned} \bar{v}_i(\vartheta) = \min_{z, h} \quad & \| [G_s^i z \leq b_s^i + S_s^i \vartheta]_+ \|_1 \\ \text{s.t:} \quad & G_h z \leq b_h + S_h \vartheta \\ & \| [G_s^j z \leq b_s^j + S_s^j \vartheta]_+ \|_1 \leq \bar{v}_j, \quad j = i + 1, \dots, n_p, \end{aligned} \quad (3.8)$$

The above problem can be interpreted as follows:

- for each priority $i = 1, \dots, n_p$ problem (3.8) returns a solution which is the optimal minimum violation $\bar{v}_i$ of the soft constraints with priority i satisfying the hard constraints and minimally violating the soft constraints with higher priority $j > i$;
- then, each optimal violation is used in (3.7) to impose a constraint on each i th set of soft constraints. In particular each i th constraint cannot be violated more than its optimal associated violation $\bar{v}_i(\vartheta)$.

It follows that the optimal solution of (3.7) must guarantee that whenever is not possible to satisfy all the soft constraints, violation of a set of soft constraints with a certain priority is considered only after that of constraints with lower priority. At the same time, as for (2.7), note that when a feasible solution exists for a given ϑ , all the optimal violations satisfy $\bar{v}_i(\vartheta) = 0$ and therefore problem (3.7) reduces to the original problem (3.1).

Hence we conjecture an extension of Falsone et al. (2023: Corollary 2) to our case stating that: under [Assumption 3.1](#), for any limit map $\bar{z}(\vartheta)$ obtained via (3.6), there exists an optimal map $z^*(\vartheta)$ of (2.1) such that

$$\bar{z}(\vartheta) = z^*(\vartheta), \quad \vartheta \in \Theta_f, \quad (3.9)$$

where Θ_f is the set of ϑ for which (3.1) is feasible.

3.3. Solution of the Relaxed Problem with Prioritized Constraints

In Chapter 2 we have recalled the structure of [Algorithm 2.1](#) of Falsone et al. 2023 to compute the optimal limit map (2.5). We now conjecture a variant to find the optimal limit map (3.6). The procedure is summarized in [Algorithm 3.1](#) below.

Algorithm 3.1: Procedure to compute $\bar{z}(\vartheta)$

```

1:  $\mathcal{P}_r \leftarrow$  optimal map  $\bar{z}(\vartheta, \boldsymbol{\mu})$  of (3.4)
2:  $\mathcal{P} \leftarrow \emptyset$ 
3: for  $k = 1$  to  $n_p$  do
4:   if  $k < n_p$  then
5:      $n = n_p - k$ 
6:     for all  $(F_i, C_i, f_i, g_i, K_i, L_i, M_i, k_i) \in \mathcal{P}_r$  do
7:       if  $f_i \leq 0$  then
8:          $\mathcal{O} \leftarrow \{j : f_i^{[j]} = 0\}$ 
9:          $F_i^0 = F_i^{\mathcal{O}}$  and  $C_0 = C_i^{\mathcal{O}}$  and  $g_i^0 = g_i^{\mathcal{O}}$ 
10:         $F_i = F_i^0, C_i = C_0^{[:,1:n-1]}, g_i = g_i^0$  and  $f_i = C_0^{[:,n]}$ 
11:         $L_i = L_i^{[:,1:n-1]}$  and  $M_i = L_i^{[:,n]}$ 
12:         $\mathcal{P} \leftarrow \mathcal{P} \cup (F_i, C_i, f_i, g_i, K_i, L_i, M_i, k_i)$ 
13:      end if
14:    end for
15:     $\mathcal{P}_r \leftarrow \mathcal{P}$ 
16:  else
17:    for all  $(F_i, f_i, g_i, K_i, M_i, k_i) \in \mathcal{P}_r$  do
18:      if  $f_i \leq 0$  then
19:         $\mathcal{O} \leftarrow \{j : f_i^{[j]} = 0\}$ 
20:         $F_i^0 = F_i^{\mathcal{O}}$  and  $g_i^0 = g_i^{\mathcal{O}}$ 
21:         $\mathcal{P} \leftarrow \mathcal{P} \cup (F_i^0, g_i^0, K_i, k_i)$ 
22:      end if
23:    end for
24:  end if
25: end for

```

Differently from [Algorithm 2.1](#), where the penalty coefficient was a scalar $\mu \in \mathbb{R}_+$, in this case we have a vector of penalty coefficients $\boldsymbol{\mu} \in \mathbb{R}_+^{n_p}$. Due to this, the procedure in [Algorithm 2.1](#) has been adapted to handle this situation. In what follows, we provide a description of it.

In the first place, at Step 1 the optimal map $\bar{z}(\vartheta, \boldsymbol{\mu})$ is computed by providing problem (3.4) as input to any available mp-QP solver and then stored as a set $\mathcal{P}_r$ of 8-tuples, where the i th element $(F_i, C_i, f_i, g_i, K_i, L_i, M_i, k_i) \in \mathcal{P}_r$ is such that $z(\vartheta, \boldsymbol{\mu}) = K_i \vartheta + L_i \boldsymbol{\mu}_1 + M_i \boldsymbol{\mu}_{n_p} + k_i$ for all $(\vartheta, \boldsymbol{\mu})$ satisfying $F_i \vartheta + C_i \boldsymbol{\mu}_1 + f_i \boldsymbol{\mu}_{n_p} \leq g_i$. Then, once the set $\mathcal{P}$ is initialized (cfr. Step 2), the *for loop* is started (cfr. Step 3). If the current μ_n , with

$n = (n_p - k + 1)$, is not the last one to process (cfr. Step 4), we set the next μ_n (cfr. Step 5) and, for each element inside $\mathcal{P}_r$ (cfr. Step 6) it is checked if the current μ_{n+1} -axis is a direction of recession for the corresponding region $\{(\vartheta, \boldsymbol{\mu}_k) : F_i^k \vartheta + C_i^k \boldsymbol{\mu}_{k+1} + f_i^k \mu_{n+1} \leq g_i^k\}$ (cfr. Step 7). If this is the case, the constraints defining the current region that do not depend on μ_n are selected (cfr. Steps 8 and 9) to update the current ith element in $\mathcal{P}_r$ (cfr. Steps 10 and 11). Subsequently the new 8-tuple is stored in $\mathcal{P}$ (cfr. Step 12), encoding the fact that $\bar{z}(\vartheta, \boldsymbol{\mu}_{k+1}) = K_i^{k+1} \vartheta + L_i^{k+1} \boldsymbol{\mu}_{k+2} + M_i^{k+1} \mu_n + k_i^{k+1}$ for all $(\vartheta, \boldsymbol{\mu}_{k+1}) \in \{(\vartheta, \boldsymbol{\mu}_{k+1}) : F_i^{k+1} \vartheta + C_i^{k+1} \boldsymbol{\mu}_{k+2} + f_i^{k+1} \mu_n \leq g_i^{k+1}\}$. When all the elements of $\mathcal{P}_r$ have been explored, this is replaced by the updated one $\mathcal{P}$ (Step 15). Once we are left with the last component μ_1 to be processed, the remaining Steps from 17 to 21 are the same of Algorithm 2.1 from Step 4 to 7. This is obvious because when the limit map is $\bar{z}(\vartheta, \mu_1)$ it depends on the scalar μ_1 only, which is the same case of the relaxed problem with no priorities.

4 | Numerical Simulations

In the following chapter we test the validity of the theoretical results conjectured in Chapter 3 by applying [Algorithm 3.1](#) on a slightly modified version of the double integrator dynamical system used as a benchmark in Falsone et al. [2023](#) and in Bemporad et al. [2000](#).

4.1. Problem Formulation

Let us consider the following discrete-time LTI dynamical system

$$x_{k+1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_k = Ax_k + Bu_k, \quad (4.1)$$

with states $x_k \in \mathbb{R}^2$ and input $u_k \in \mathbb{R}$. The objective of the finite-horizon optimization control problem is to regulate the first state of system (4.1) to the origin while minimizing the control effort and enforcing constraints on both the actuation and the states. Since we need at least two state constraints to have different level of priorities, differently from Falsone et al. ([2023](#)), the additional requirement that the first state should stay within the interval $[-8, 8]$ is imposed, besides having that the second state and the control input both must lie in the interval $[-1, 1]$ as required in Falsone et al. [2023](#). The resulting formulation of the problem is

$$\begin{aligned} \min_{\{u_t, x_t\}_t} \quad & \sum_{t=k}^{k+N-1} x_{k+1}^T Q x_{k+1} + u_k^T R u_k \\ \text{s.t.:} \quad & -8 \leq x_t^{[1]} \leq 8 & t = k+1, \dots, k+N \\ & -1 \leq x_t^{[2]} \leq 1 & t = k+1, \dots, k+N \\ & -1 \leq u_t \leq 1 & t = k, \dots, k+N-1 \\ & x_{t+1} = Ax_t + Bu_t & t = k, \dots, k+N-1, \end{aligned} \quad (4.2)$$

with $Q = \text{diag}([1 \ 0])$ and $R = 0.1$. The returned optimal control law is applied in

a receding-horizon fashion: at a generic discrete-time instant k we solve (4.2) for the current state x_k , we feed the optimal policy u_k^* to the system, we repeat the process at the next time instant $k + 1$ and so on.

The problem can be easily converted to an mp-QP having the same structure as (2.1): we use the dynamic equation (4.1) to express each x_t as a function of x_k and u_s , $s = k, \dots, t - 1$, $t = k, \dots, k + N$. Therefore we set $z = [u_k \ \dots \ u_{k+N-1}]^T \in \mathbb{R}^N$ and $\vartheta = x_k \in \mathbb{R}^2$, then we build G_h and b_h to model the actuation constraints $u_t \in [-1, 1]$ for $t = k, \dots, k + N - 1$, with $S_h = 0$ since they do not depend on ϑ , while G_s, b_s and S_s model the state constraints $x_t^{[1]} \in [-8, 8]$ for $t = k + 1, \dots, k + N$ and $x_t^{[2]} \in [-1, 1]$ for $t = k + 1, \dots, k + N$. From now on we will refer to the problem formulations introduced in the previous chapters as instances related to this example.

4.2. Software Tools

The problem is formulated in Matlab using *Yalmip*, a software tool which allows high-level operations for modeling optimization programs. In this way it is easier to implement the formulation of the multi-parametric quadratic program: the constraints and the objective function are defined as variables according to the MPC formulation (4.2) and then given as input to an auxiliary optimization solver. The one used here is the latest version of the *Multi-parametric Toolbox* (MPT3): a software tool developed for Matlab to solve parametric optimization programs related to their optimization control problem counterpart, Herceg et al. (2013).

4.3. Solution without Prioritized Constraints

The first experiment we want to conduct is to feed MPT3 with the Yalmip code for the parametric program (2.1), which returns the optimal map $z^*(\vartheta)$. This computation is required only once and can be used to control the system by setting the control law as $u_k^*(x_k) = z^{*[1]}(x_k)$, which is then applied at each k .

If we run Algorithm 2.1 using as time horizon $N = 3$, we obtain the limit map $\bar{z}(\vartheta)$, in full accordance with Falsone et al. (2023: Theorem 1), defined over the regions partitioning the state space represented in Fig. 4.1. From this figure it is possible to observe that the regions do not intersect each other, they are polyhedral, and, furthermore, the entire Θ -space is covered (which coincides with $\mathbb{R}^2$ being $S_h = 0$), as expected from Falsone et al. (2023: Corollary 1), thus confirming that problem (2.3) admits a solution for each $\vartheta \in \mathbb{R}^2$.

The next result we want to verify is the equality of the relaxed problem with the non-relaxed one within the feasible region. In order to do this we show both the partitions

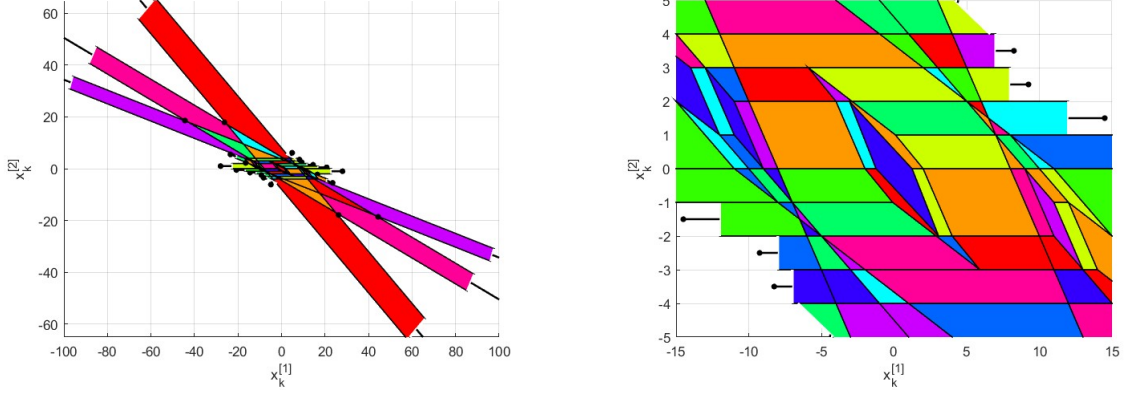


Figure 4.1: Polyhedral regions partitioning the state space for which problem (2.3) is feasible on the left and its zoomed image on the right. Each color defines a different region and black whiskers indicate directions in which each region extends to infinity

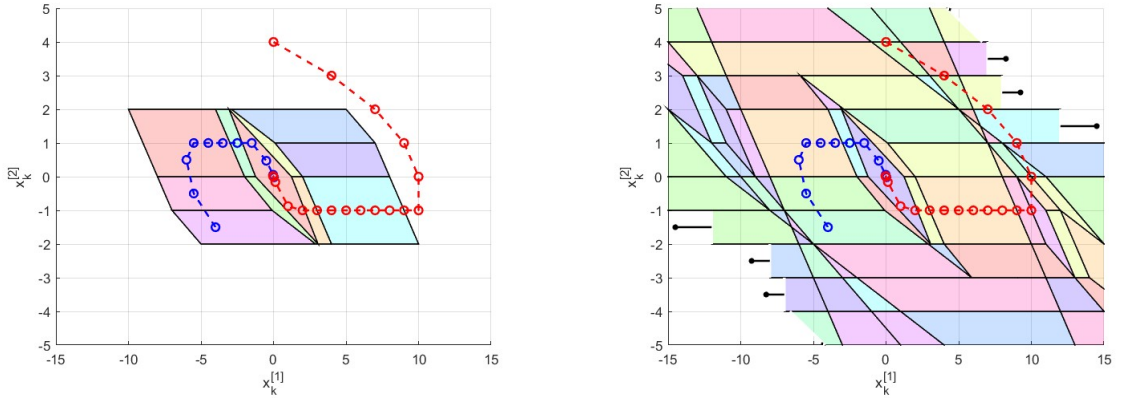


Figure 4.2: Partition induced by the optimal solution z^* of (2.1) (left) and by $\bar{z}$ in (2.5) (right). Dashed lines with circles refers to the 20-step long trajectory of system (4.1) initialized at $x_0 = [-4 \ -1.5]^T$ (blue) or $x_0 = [0 \ 4]^T$ (red) generated by applying either the control law $u_k(k) = z^{*[1]}$ (left) or $u_k(x_k) = \bar{z}^{*[1]}(x_k)$ (right), for $k = 0, \dots, 19$.

of the state space defining $z^*(\vartheta)$ and $\bar{z}(\vartheta)$ in Fig. 4.2.

A first indication is given by the same shape the regions share when the states belong to the feasible set of states $\vartheta \in \Theta_f$. It is possible to provide a further validation initializing the state of the system at $x_0 = [-4 \ -1.5]^T$ (which is feasible) and comparing the trajectory generated by applying the control law $u_k^* = z^{*[1]}(x_k)$ (Fig. 4.2 left, blue dashed line) for each $k = 0, \dots, 19$ with the one obtained using $u_k^* = \bar{z}^{*[1]}(x_k)$ (Fig. 4.2 right, blue dashed line) for each $k = 0, \dots, 19$. From the pictures it is straightforward to notice that the two trajectories are described by the same line and thus the claim of Falsone et al. (2023: Corollary 2) is verified.

The second test is aimed at verifying how the system behaves when its state is outside the feasible region $\vartheta \notin \Theta_f$. Hence this time we initialize the system at the state $x_0 = [0 \ 4]^T$ and compute two trajectories: the first (Fig. 4.2 left, red dashed line) applying the control action $u_k(x_k)$ which minimizes the 1-norm violation of the state constraints when the system state is outside the feasible region and $u_k(x_k) = z^{*[1]}(x_k)$ when the state is inside, as prescribed by (2.7) and (2.8); the second (Fig. 4.2 right, red dashed line) using the control law $u_k(x_k) = \bar{z}^{[1]}(x_k)$, both for $k = 0, \dots, 19$. The obtained trajectories are equal, validating thus the results stated by Falsone et al. (2023: Theorem 2).

4.4. Solution with Prioritized Constraints

Let us now consider (3.1) in which the two state constraints have different priorities. We apply Algorithm 3.1 to solve the prioritized problem (3.2) and obtain the limit maps $\bar{z}_{p_1}(x_k)$ and $\bar{z}_{p_2}(x_k)$ in (3.6), where the subscripts p_1 and p_2 indicates that we are prioritizing the constraint on $x_k^{[1]}$ over that on $x_k^{[2]}$ or vice versa, respectively. From the partitions induced by $\bar{z}_{p_1}$ (left) and $\bar{z}_{p_2}$ (right) in Fig. 4.3 we can notice that:

1. both partitions induced by the limit map cover the entire Θ -space and are polyhedral;
2. inside the feasible regions they have the same shape of the partition in Fig. 4.1;
3. some regions (highlighted with a darker shade) slightly differ from each other and from the ones of partition in Fig. 4.1.

The last consideration is a first suggestion that the maps $\bar{z}(x_k)$, $\bar{z}_{p_1}(x_k)$ and $\bar{z}_{p_2}(x_k)$ have different values for the same x_k inside specific regions, meaning that performing an ordered sequence of limits along each component of $\mu \rightarrow \infty$ produces a result which is not only different from doing just the limit along a scalar $\mu \rightarrow \infty$ penalizing both the violations, but it depends also on the order of the limits.

The second test wants to prove that the system behavior inside the feasible region Θ_f is the same for all the three limit maps. We can verify it initializing the state of the system at $x_0 = [-4 \ -1.5]^T$ and applying the control laws $u_k = \bar{z}_{p_1}^{[1]}(x_k)$, $u_k = \bar{z}_{p_2}^{[1]}(x_k)$ and $u_k = \bar{z}^{[1]}(x_k)$, for each $k = 0, \dots, 19$. The resulting trajectories are illustrated in Fig. 4.4a, Fig. 4.4b and Fig. 4.4c respectively, as blue dashed lines. As it can be seen they are the same, thus confirming what conjectured by (3.9).

The other test performed is aimed at investigating if the state trajectories behave differently for each case when the system starts outside the feasible region and how the penalization of the constraints is carried on in each of them. The procedure to do this is

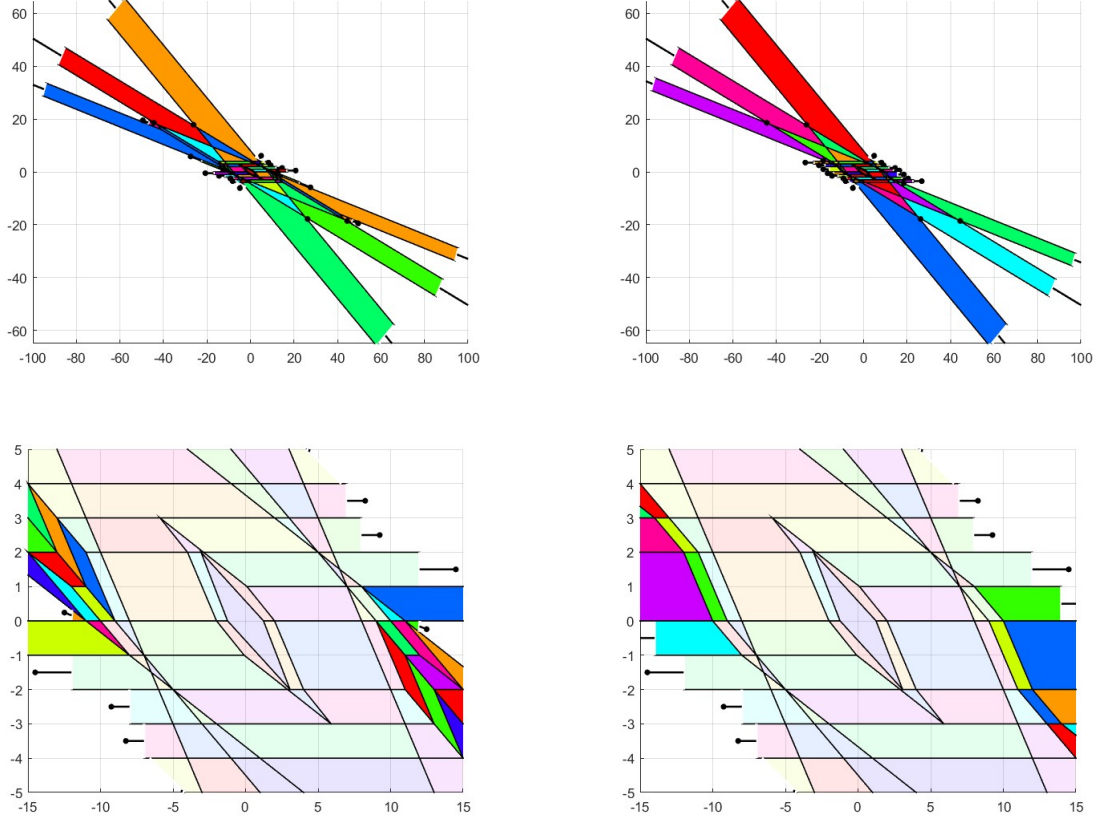


Figure 4.3: The two pictures on the tops are the partition induced by $\bar{z}_{p_1}$ (left) and $\bar{z}_{p_2}$ (right) of (3.2), while the ones on the bottom are the same partitions zoomed, where the different regions are highlighted by an higher intensity of their color

the same as before: we initialize the state of the system at $x_0 = [14.5 \quad -3]^T$ and apply the control actions $u_k = \bar{z}_{p_1}^{[1]}(x_k)$, $u_k = \bar{z}_{p_2}^{[1]}(x_k)$ and $u_k = \bar{z}^{[1]}(x_k)$, for each $k = 0, \dots, 19$. The obtained trajectories are reported in Fig. 4.4a, Fig. 4.4b and Fig. 4.4c respectively, as red dashed lines.

We also report in Table 4.1, Table 4.2 and Table 4.3 the values assumed by the state and the input along the trajectories in Fig. 4.4a, Fig. 4.4b and Fig. 4.4c respectively, for the first five time steps $k = 0, \dots, 4$. Let us comment the numbers reported in this tables:

1. it is possible to notice from the first control action $u_k(x_k) = \bar{z}_{p_1}^{[1]}(x_k) = -0.5$ at $k = 0$ in Table 4.1 that the reduction of the violation of the constraint on $x_k^{[1]}$, which has higher priority, comes at the expenses of an increase in the violation of the constraint on $x_k^{[2]}$. Indeed at the next time step $k = 1$ the second state $x_k^{[2]}$ is further away from the interval $[-1, 1]$ imposed by the constraint while $x_k^{[1]}$ gets closer to constraint satisfaction. This increase of $x_k^{[2]} = -3.5$ at $k = 1$ allows to decrease $x_k^{[1]}$ by a larger amount such that at $k = 2$ we have $x_k^{[1]} = 8$ and the constraint on the

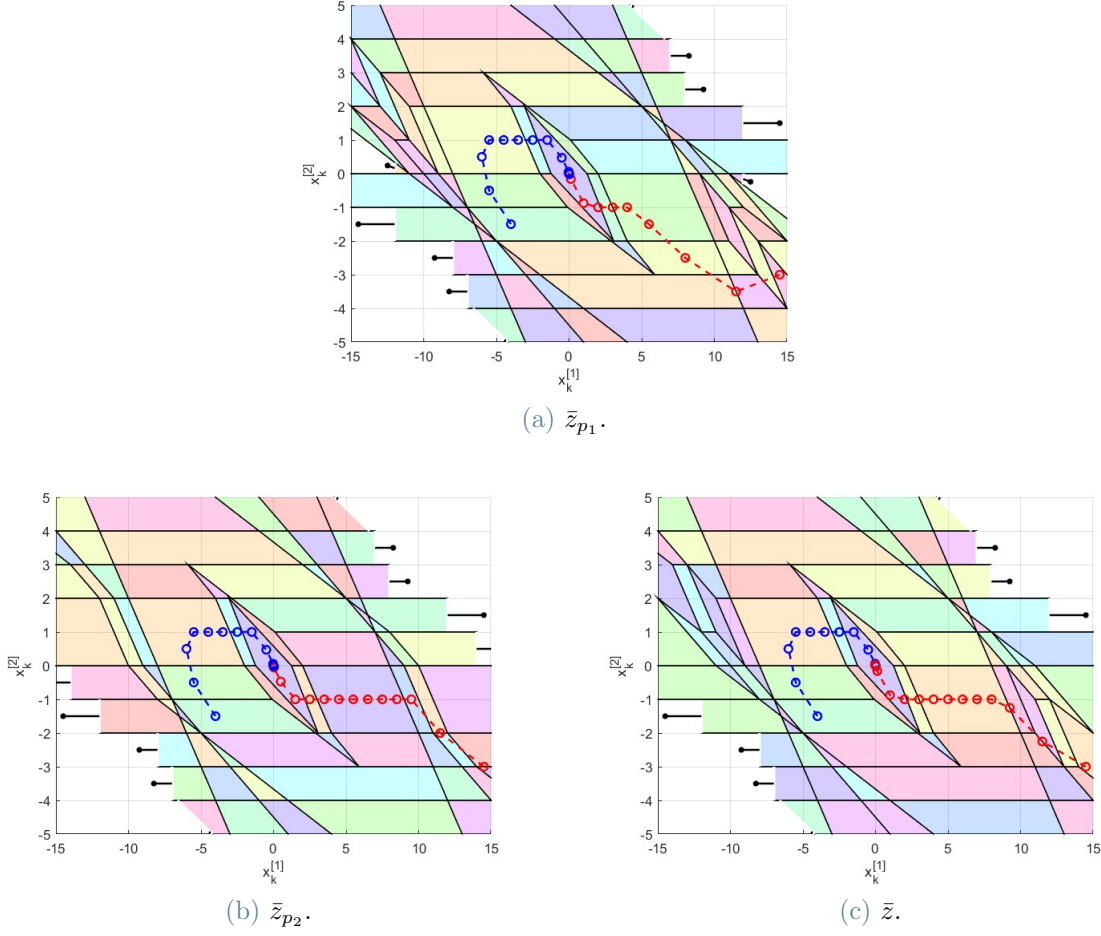


Figure 4.4: Trajectories generated by $u_k = \bar{z}_{p_1}^{[1]}(x_k)$, $u_k = \bar{z}_{p_2}^{[1]}(x_k)$ and $u_k = \bar{z}^{[1]}(x_k)$ when the system is initialized either at $[-4 \quad -1.5]^T$ (blue dashed line) or $[14.5 \quad -3]^T$, for $k = 0, \dots, 19$.

first state is satisfied, while the constraint on the second state is satisfied at $k = 4$;

2. from Table 4.2 it can be seen that the first control action $u_k(x_k) = \bar{z}_{p_2}^{[1]}(x_k) = 1$ at $k = 0$ goes in an opposite direction respect to the one in Table 4.1. With this choice at $k = 1$ the second state $x_k^{[2]} = -2$ is now decreased as much as possible, at the price of slowing the decrease of $x_k^{[1]}$ towards $[-8, 8]$, which depends on the value of $x_k^{[2]}$. At $k = 2$ the constraint on the second state is satisfied while the one on the first state is satisfied at $k = 4$;
3. in Table 4.3 we can see that the first control action $u_k(x_k) = \bar{z}^{[1]}(x_k) = 0.75$ at $k = 0$ is in between the ones in Table 4.1 and Table 4.2. Applying $u_k = \bar{z}^{[1]}$ for $k = 1, \dots, 2$ the system ends up to the state $x_k = [8 \quad -1]^T$ at $k = 3$. Hence this solution is a compromise between the ones taking into account priorities. No one of constraints

is satisfied before step $k = 3$, but both of them are satisfied at $k = 3$, contrary to the two previous cases where satisfaction of both constraints happens at $k = 4$.

	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$
$x_k^{[1]}$	14.5	11.5	8	5.5	4
$x_k^{[2]}$	-3	-3.5	-2.5	-1.5	-1
$\bar{z}_{p_1}^{[1]}(x_k)$	-0.5	1	1	0.5	0

Table 4.1: The first two rows and the third one contain respectively the values of state variables x_k and the input u_k along the trajectory in Fig. 4.4a generated by $\bar{z}_{p_1}^{[1]}(x_k)$, for $k = 0, \dots, 4$.

	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$
$x_k^{[1]}$	14.5	11.5	9.5	8.5	7.5
$x_k^{[2]}$	-3	-2	-1	-1	-1
$\bar{z}_{p_2}^{[1]}(x_k)$	1	1	0	0	0

Table 4.2: The first two rows and the third one contain respectively the values of state variables x_k and the input u_k along the trajectory in Fig. 4.4b generated by $\bar{z}_{p_2}^{[1]}(x_k)$, for $k = 0, \dots, 4$.

	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$
$x_k^{[1]}$	-14.5	11.5	9.25	8	7
$x_k^{[2]}$	-3	-2.25	-1.25	-1	-1
$\bar{z}^{[1]}(x_k)$	0.75	1	0.25	0	0

Table 4.3: The first two rows and the third one contain respectively the values of state variables x_k and the input u_k along the trajectory in Fig. 4.4c generated by $\bar{z}^{[1]}(x_k)$, for $k = 0, \dots, 4$.

Starting from $[13.5 \quad -3]^T$ the system reaches either the state $x_k = [4 \quad -1]^T$ when controlled by $u_k = \bar{z}_{p_1}^{[1]}$, $x_k = [7.5 \quad -1]^T$ when $u_k = \bar{z}_{p_2}^{[1]}$ or $x_k = [7 \quad -1]^T$ if we use $u_k = \bar{z}^{[1]}$, at $k = 4$. The only control strategy who managed to obtain a zero violation of both constraints earlier is the one penalizing both of them in the same way, $u_k = \bar{z}(x_k)$.

The others violate the constraint which has lower priority, leading us to conclude that: considering priorities allows to satisfy earlier the prioritized constraint, but this may penalize the satisfaction of the other constraint, which needs another time step to be satisfied.

A further confirmation of the validity of our result can be obtained by feeding the system, initialized at $[14.5 \quad -3]^T$, with $u_k = \bar{z}_{p_1}^{[1]}$ or with $u_k = \bar{z}_{p_2}^{[1]}$, obtained by solving (3.7), for $k = 0, \dots, 2$ (until the state enter in the feasible region), respectively with

$$\begin{aligned}
\bar{v}_2(\vartheta) &= \min_{z,h} \quad \|[G_s^2 z \leq b_s^2 + S_s^2 \vartheta]_+\|_1 \\
\text{s.t: } &G_h z \leq b_h + S_h \vartheta \\
&\|[G_s^1 z \leq b_s^1 + S_s^1 \vartheta]_+\|_1 \leq \bar{v}_1 \\
\bar{v}_1(\vartheta) &= \min_{z,h} \quad \|[G_s^1 z \leq b_s^1 + S_s^1 \vartheta]_+\|_1 \\
\text{s.t: } &G_h z \leq b_h + S_h \vartheta
\end{aligned} \tag{4.3}$$

or

$$\begin{aligned}
\bar{v}_1(\vartheta) &= \min_{z,h} \quad \|[G_s^1 z \leq b_s^1 + S_s^1 \vartheta]_+\|_1 \\
\text{s.t: } &G_h z \leq b_h + S_h \vartheta \\
&\|[G_s^2 z \leq b_s^2 + S_s^2 \vartheta]_+\|_1 \leq \bar{v}_2 \\
\bar{v}_2(\vartheta) &= \min_{z,h} \quad \|[G_s^2 z \leq b_s^2 + S_s^2 \vartheta]_+\|_1 \\
\text{s.t: } &G_h z \leq b_h + S_h \vartheta
\end{aligned} \tag{4.4}$$

as instances of (3.8). Then we apply $u_k(x_k) = z^{*[1]}(x_k)$ of (2.1) for the remaining time steps $k = 3, \dots, 19$. The resulting trajectories are depicted in Fig. 4.5 and as it can be seen they are equal to the ones in Fig. 4.4a and Fig. 4.4b (dashed red lines) generated respectively by $\bar{z}_{p_1}^{[1]}$ and $\bar{z}_{p_2}^{[1]}$ obtained from Algorithm 3.1. This suggests that also our conjecture on the relationship between problem (3.2) and (3.1) is correct.

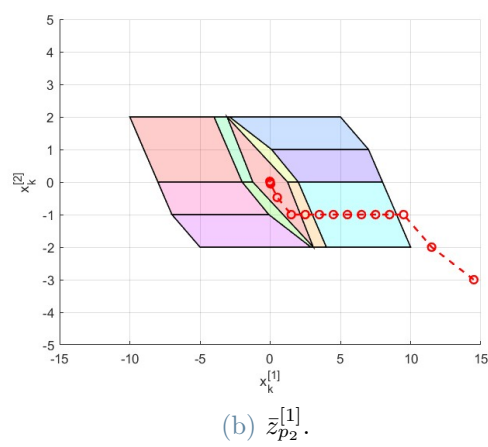
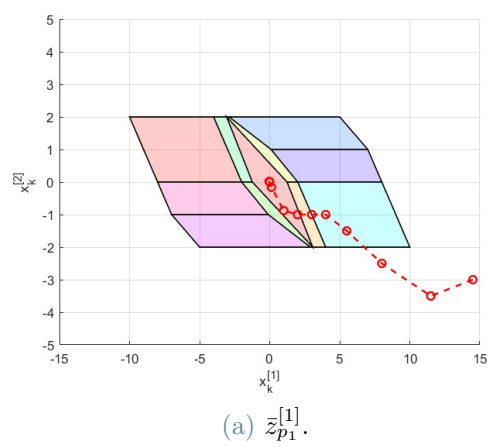


Figure 4.5

5 | Conclusions

We started recalling the idea behind optimal control and showing how the infinite horizon optimal control problem can be approximated by using a finite time horizon. In so doing the new problem can be solved either as a quadratic program using the standard MPC, or, by treating the states as parameters, as a multi-parametric quadratic program using the explicit version of MPC.

In the second Chapter we introduced the infeasibility issue and illustrated that this can be counteracted relaxing the soft constraints on the states. One way to do so is adding to the objective function of the mp-QP formulation a scalar weight penalizing the 1-norm soft constraints violation. The solution of this new problem, under suitable assumptions, has the property to converge to a limit piece-wise affine map independent of the penalty weight as it grows to infinity. Moreover, this map coincides with the solution of the original hard problem when the states are inside the feasible region.

In Chapter 3 we resumed the same relaxed problem, this time classifying each soft constraint according to a different priority level, and then formulated a new mp-QP where each constraint violation with a different assigned priority is penalized by a corresponding scalar coefficient. The rest of the chapter aims to conjecture that the solution of this new problem with prioritized constraints shares the same properties of the one obtained solving the relaxed problem. In particular, performing a sequence of limits (equal to the number of priority levels and reflecting their descending order) along each coefficient approaching infinity, the solution converges to a limit map independent of the coefficients which is the same solution of the original hard problem whether the states belong to the feasible region. But most importantly, the obtained limit map is such that the constraints violations with higher priority are firstly minimized over the ones with lower priority.

In the last chapter we firstly validated the theoretical results of Chapter 2 testing the algorithm on a double integrator system. After that, we compared these results with the ones obtained by applying the algorithm of Chapter 3 on the same dynamical system, one time prioritizing the constraint on the first state over the second one, and once vice-versa. From the generated state trajectories it is evident that their evolution is different in all three cases and in accordance with what conjectured in Chapter 3.

A natural outlook of this thesis would be to rigorously prove the conjectures in Chapter

3 and test the approach on a realistic case study to assess how constraint prioritization can affect the optimal control strategy when the state lands into the infeasible region.

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